

Matchmeup

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Basics: What is Matchmeup?

I plan to develop, design, test, and market a web tool and website to encourage “partnering” with a wide variety of applications.

1. A “Founder user” can set up a “Partner exchange” page on our website.
 2. The Founder can enter a list of names and invite these people (with an optional password logon) to use the page.
 3. These users can login, fill out a profile if desired, and view the list of other names on the exchange.
 4. They can then select one or more of the other listed users whom they would wish to partner with.
- Depending on how the founder sets up the exchange, the users can make their choices either
 - with “Conditional Anonymity”, where the other person will only be told that they were chosen if they also choose the first user,
 - or with “Full Revelation” – where the user is informed who chose him in any case, or they can choose on a case-by-case basis.
 - There is some precedent for this; dating websites have offered these possibilities in certain contexts, but the tool and service I propose is new.

The site will be developed to give the founder a great deal of flexibility to set other restrictions and add other features to this interface. For example, the Founder could impose a limit on the number of choices one could make, could impose a cost to make each choice (e.g., a £1 donation to charity per choice), and may be allowed to partition the “matching market” into groups that are allowed to match (under some limitations). However, all of this will need to be made plain to the other users and we will not permit any deception or invasions of privacy. (More on this later.)

To get people to use the site, the Founder (or founders) will also be able to offer **prize incentives**. For example, the founder will be able to pay for a £20

pound Amazon gift certificate, and tell all the invited users, that one user will be randomly awarded this gift certificate. We will also consider allowing the Founder to charge users if they want to use his or her page, perhaps asking for a charitable donation. Users who have a “match” – either they chose someone who also chose them in the conditionally anonymous treatment, or someone chose them in the Full Revelation treatment – can be informed by e-mail, and they can log on to see their matches.

This site will have numerous uses: academic, administrative, commercial, and personal.

Trials and launch plan, proposed use cases

Trial 1: Interdisciplinary Research and Knowledge Exchange Partnering at the University of Exeter and associated conference

A likely first application will be a page that University of Exeter sets up inviting researchers find potential research partners, particularly in other disciplines, and to find matches between academic researchers and private firms, entrepreneurs, nonprofits, and public organisations. This will be particularly applicable to research clusters in the Business School and attendees at conferences.

In some cases, merely encouraging face-to-face meetings may allow colleagues to reveal their potential interest slowly and conditionally, lessening the risk of a severe Loss of Face from a “desperate bid”. This may help explain Boudreau et al (2012); who found that an exogenously assigned brief face-to-face interaction between local scientists led to significant increases in their probability of collaborating. This logic may have motivated the “Speed Dating for Academics” the REO held in March 2008. However, we imagine that – because of reputation concerns, the primal fear of “losing face”, and cultural shyness – allowing truly conditionally anonymous choices might do even better, and we would like to help test or trial this.

I will propose that University host one of the pages, as a founder user. There may be separate pages for faculty and graduate students, or this could be done through filtering. The University will offer a small prize, randomly chosen, for one member of faculty, one graduate student, one entrepreneur, and one private enterprise, nonprofit, or public organisation who posts a complete profile on the page. However, to get things going, the University may choose to include a very basic profile for all researchers who do not opt out on the site. Researchers, organisations and entrepreneurs, can then log on, post their profile or not do so, and “tick” the names of the others they would like to collaborate with. The site could also be made more sophisticated allowing people to enter levels of interest, and specify areas of participation and collaboration, e.g., collaboration

in grant writing, collaboration on a paper, and collaboration in data sharing. We may also allow people to make suggestions for which other researchers say think might be interested in collaborating.

It is likely to be more successful, however, if we don't pose it in terms of a commitment to "collaborate" – instead just a commitment to meet to discuss collaboration. I believe that even this will be more successful if we allow a conditional anonymity option on the site. However, I would like to test this, as it is also possible that not making this an option would be more successful, as people may be more likely to reciprocate when they get an offer even though they would not have proposed an offer to the other party themselves. Thus, if possible, I would like to propose a controlled randomization of the conditional anonymous offer option, perhaps at a group level.

I may also introduce and promote this at conferences where I am presenting my paper, and host/advertise it on my web site, davidreinstein.wordpress.com/

Trial 2: Student study and project partnering at the University of Exeter

A second early application and trial will encourage students to choose study partners within their class or lecture. For example, in large lecture classes, the GTA can be encouraged to set up a site for each class. They will put the names of their students on the site, unless they choose to opt out, and each student can log in and select which other student(s) they might wish to be study partners with. To test the effectiveness of the distinct ways of organizing the matching, I propose a balanced randomization of the matching designs at the lecture level and then again at the class level. Clearly, this will have to be negotiated with the departments and with the module supervisor. Again, I expect there to be more offers, and perhaps more matches, when these can be made under conditional anonymity, as this will reduce people's fear of losing face and reduce their fear of hurting their own reputations. However, there are also arguments that not allowing this would be more effective, as people again may reciprocate when they're asked even if they would not have made an offer themselves, for various reasons. As in the researcher-matching context, if we run a companion survey, we will be able to quantify the effects of each type of site, both in terms of the number of matches made, and in the success of these (number of meetings, perceived success of interactions, marks achieved).

Trials 3: Collaborative Consumption

This site will serve as a research tool for my project with Thomas Gall, "Losing Face and Reputation in Matching Environments." We are pursuing field experiments and trials involving community building, sharing information with

neighbours and “Collaborative Consumption.” This may build on collaborations with Nesta and the Collaborative Consumption Hub. Our brief mimeo on this: “Risk-free partnering exchanges: combatting shyness and promoting collaborative consumption” is attached.

We also have plans (and have been granted ethical approval) for an experiment specifically asking neighbours if they would like to exchange emails with one another. This application may prove useful in this experiment.

Trials 4: Work with social groups

We will advertise and promote this tool for use by community, church, school, and social groups to encourage partnerships, romantic and otherwise. We may start by asking for the promotion and support of the Exeter SU and the SU at other universities.

Launch of Web site for public use

After appropriate trials have been done and hard evidence gathered, and after the site has been well built and proved robust in the above trials, we plan to launch the site (or a branch of the site) for public use, and for commercial gain (perhaps under a non-profit umbrella). This is further discussed below. At this point (and leading up to this) we may look for outside grant funding and private venture capital.

Expertise

David Reinstein is a Senior Lecturer in Economics at the University of Exeter Business School. He has conducted and published research on behavioral economics, consumer behavior, and charitable giving and altruism for over a decade. He is enthusiastic about facilitating controlled field experiments as well as less formal trials and pilots. He has organised and run several laboratory and field experiments, including a speed dating event and a Valentine’s day website allowing students and staff to exchange cards under anonymous, revealed, or conditional revelation environments. This site was designed by a team of academics with help from outside consultants, and involved a secure website, a graphically appealing design, an interface with a SQL database, and an interactive user interface involving JavaScript and php. David’s research has involved analysis of a wide variety of data, including data stripped from web sites using “crawlers” and other tools. He is also actively pursuing relationships and field experiments with private and public sector organizations. In previous employment at the Haas Business School (Berkeley) he has worked with Professor Pablo Spiller to analyse social interaction on the internet, sharing of information

goods, and other topics in “Internet Economics.” He has worked on antitrust and intellectual property issues for LECG Consulting in California. David has pursued a strong Knowledge Exchange agenda, working on field experiments on Payroll Giving with the Cabinet’s Behavioural Insights team, and as part of a NESTA-funded project extending the Brixton Pound to the Lambeth County Council payroll. Each of these experiments engage with firm-specific processes, databases, and IT systems.

His recent paper (Gall and Reinstein) takes a formal analytical perspective, as well as drawing on a range of informal evidence. We find “loss of face” (LoF) may be explained by a reputation mechanism, and it may also enter directly into the utility function. In either case, this can lead to fewer offers and inefficiency in the context of bilateral matching problems, e.g., the marriage market, friendship-formation and activity partnering, certain labour markets, and international negotiations. In our model, a larger amount of LoF leads high types to fully leave the market, and makes the lower types become “reverse snobs,” also reducing welfare. Where LoF harms welfare, when only one side is vulnerable to LoF, welfare can be improved by letting this party be the second-mover. More generally, we propose a “Conditionally Anonymous Environment,” i.e., a mechanism that only reveals when both parties say “yes.”

Overall market potential

Wide applicability

The site may be used by any social, business, government, or academic organisation looking to encourage its members to overcome their shyness and “partner up”. For example, we can envision this being wildly popular among teenagers with a “secret crush”, who will sponsor a site for their class in order to learn if their feelings are requited, while saving face in case the feelings are not mutual. At another extreme, a legal “headhunting” firm may sponsor a site to find potential matches between lawyers and firms.

Matchmaking and partnering consulting and intermediation

This website can be used as a research tool, in ways such as the one I’ve mentioned. This tool may be licensed to other researchers. More importantly, the understanding gleaned about how people make matches in various sectors of life, will not only be interesting, it may lead to fruitful consulting possibilities. Corporations, universities, dating services, and marketers, may seek consulting about the best ways to foster collaboration in their environments, and what factors will encourage successful partnering. For example, a speed dating agency

may gain from this research, and may seek advice about how to match people at tables, and how to structure the environment in which they're able to make offers to one another. There is no consensus on the "best" way to do this. Quickly examining eight speed-dating agencies (Speed Dater, Click Singles, Slow Dating, X Factor Dates, Ditch or Date?, When the Music Stops, Fast Love, and Date & Dash) we found four reveal only mutual "yes,yes" matches (one of these also reveals the number of ticks you got), two reveal anyone who ticked you, one also reveals "warm; yes, maybe" matches, and one allows you to pay a fee to see who ticked you even though you didn't tick them (a "second chance"). Three of these also allow "friendship matches" of some form.

This research will also be of interest to other groups trying to encourage socialization, such as cruise directors and churches. Local and regional business promotion groups may also have something to learn about the ways to foster partnerships and bring businesses together in starting new enterprises. To the extent face-to-face meetings are required, we could facilitate these at the University of Exeter perhaps leveraging the FEELE lab and making use of the Innovation Centre.

Web site and services

This website itself may be marketable. There are many fee-paying models, but a natural way to monetize this is through the prizes that individuals looking to kickstart matching pages will offer. If we charge even a small percentage of these prizes in order to administer them, if the site comes used widely in classrooms universities social groups and businesses, these small payments will add up significantly.

The ability to make money off of this in the medium and long run will depend in part in our ability to ward off copycat competition. Our site may not be difficult to create in a technical sense. However, there will be a certain amount of network externalities and network economies of scale and a related first-mover advantage. More importantly, we may be able to use intellectual protection to protect our idea from competition. This process is both useful and nonobvious, and it may be able to secure a patent

Perhaps our most important strategic advantage is our credibility. One impediment to many "matchmaking" applications, especially applications on Facebook like "Are You Interested", is that they were (or were perceived to) manipulate the results, trying to lure potential users by inventing someone who "liked" them. They also relentlessly nagged people to invite their friends, and were seen as intrusive. The administrators were unknown, and little data protection was offered. Users must be confident that neither the Founder, nor other users will learn their choices (unless they are matched), and in many circumstances the Founder will also want to preserve his or her anonymity. Credibility is essential for a partnering site to work; neither startups nor for-profit social networking

sites are widely trusted. As a trusted non-profit academic organization, the University of Exeter will be able to guarantee careful data protection and truthful reporting of results. This credibility will also be important in getting Founders to sponsor a prize; they must trust us to randomly and fairly choose the prize winner from eligible users, and send him or her the prize.

Income generation

There are too many uncertainties to give a confident estimate of the potential profits. The potential forms of income are described in the previous section. However, we offer some background statistics on relevant industries as a general guide.

Online dating: This industry earned an estimated \$2 billion in revenue in the US in 2012 (IBISWorld report, <http://www.ibisworld.com/industry/default.aspx?indid=1723>). According to a web source (methodology not reported) for the UK “in 2011, the industry turnover amounted to 168.5 million gbp” <http://www.leadingdatingsites.co.uk/press/online-dating-market-2011-2012-uk.pdf>

Consider also

- Research grant income
- Researchgate and Academia.edu revenue and market capitalization
- Revenue from selling search and research tools to Universities and Consulting/Research firms

Competition analysis

After an extensive search (get frontrunners on this) we could not find any comparable tool on the internet. There is no tool that allows someone to set up a secure and ‘rejection free’ matching environment and to offer prizes to get people to participate.

In the dating world, the most comparable and well-known site is Tinder, and there are others following a similar protocol. Conditional anonymity is often an option, although the sites typically are more likely to *show* you people who already clicked (or ‘swiped right’). Note that with thousands of eligible members there is a cost: if A expresses conditionally anonymous interest there is no guarantee that B will even see A’s profile. In contrast, for our site, where Founders will sponsor a site for a limited group, this will be less of an issue; in a small group everyone will “consider” everyone else.

Funding (some rough numbers from a previous request; we could do much with less money)

£7125 plus overhead if necessary, plus costs mentioned at bottom if possible and necessary

- £3225 to hire one or more skilled programmers and web developers/designers (estimated: 150 hours at Grade 9, spinal point 40, or hiring outside labour), preferably a student at UoE, to help us develop and create a flexible online tool, as well as a specifically tailored site for the university projects (trials 1 and 2). The programmers will also pursue more involved search and matching algorithms.
- £300 to hire a designer to improve the look and feel of the web pages and help prepare promotional material.
- £300 for web hosting, database access, and domain name registration.
- £300 for prize incentives to use the site in trials 1 and 2 (we will ask the university and faculties to supplement these prizes).
- £700 for classroom based trials of key design points at the Essex Social Sciences Experimental facility
- £1350 for travel expenses: £300 = Six trips to London to work with the Cabinet Office, Nesta, and other potential collaborators. £150 = Five trips to local schools and businesses, to discuss further potential trials of the site. £900 for travel to meetings and conferences in the UK and abroad
- £350 for advertising and promotional expenses for trials and to garner interest in the site.
- £300 to hire 100 users on Mechanical Turk to quality-check the site.
- We may also need support from University intellectual property, legal, and business development experts, and there are potential incorporation and patent application costs.